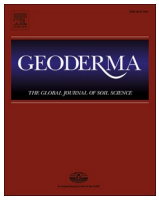
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Mapping soil property classes over a large territory with multiple soilscapes by digital extrapolations of legacy detailed soil maps: A case study in Karnataka -South India

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ABSTRACT

Using detailed soil maps to calibrate DSM models could be an alternative to point observations, as they would account for local soil patterns more accurately than the sparse sets of soil profiles classically used in broad-scale DSM applications. However, the detailed soil surveys are most often scarce on large territories, which generates clustered calibration sets that may not represent the whole unmapped area. It is therefore important to delineate extrapolation areas that have soil-landscape relationships sufficiently similar to those of the soil map perimeter.

We developed a DSM approach for mapping soil property classes (depth, texture and stoniness) over a large part (156,499 km^2) of Karnataka state, South India. We used a sparse set of 91 soil maps of micro-watersheds (464 km^2 of soil mapped areas) collected from recent land inventory programmes. Soilscape distances between soil maps were first defined by measuring the differences between soil property class distributions for each couple of soil-mapped micro-watersheds. A predictive model (random forest) that can estimate these 'ground-truth' soilscape distances was then calibrated by using as covariates the differences of distributions and variograms of soil covariates (e.g. relief, climate, remote sensing data and small-scale soil maps), as well as the geographical distance between micro-watersheds. Soilscape distances were then used to select the appropriate DSM model for predicting soil property classes at each location (i.e. the model calibrated with the map of the closest micro-watershed). Soilscape distances served also to delineate extrapolation areas around existing soil maps in which soil property classes can be predicted with the highest accuracy and lowest predicted uncertainty.

Using a leave-one-micro-watershed-out evaluation approach, We found that a single model calibrated onto the entire set of soil maps successfully predicted the texture and stoniness classes of soils over an extrapolation area covering 7% of the entire study area. Accuracies of 94% and 90% were obtained for texture, and stoniness, with respective predicted uncertainties of 6% and 7%. However, lower accuracy (57%) and higher uncertainty (31%) were obtained for predicted soil depth classes. Using multiple DSM models, each selected from soilscape distances, did not improve upon these results.

This exploratory study paves the way for a possible hybrid approach to mapping soils across large territories. This approach would combine conventional soil surveys for detailed mapping of soil properties with digital soil mapping to extrapolate detailed soil maps. Digital soil mapping sampling techniques should also be employed in the future to select the locations of further detailed soil maps for mapping the target territory in an optimal way, thereby extending the extrapolation area while reducing survey costs.

1. Introduction

The initial promise of digital soil mapping (DSM) was to combine large area coverage, fine spatial resolution and high accuracy of soil predictions to enable users to make decisions based on sufficient soil knowledge (Sanchez et al., 2009). The DSM operational projects launched on a global scale fully met the first two criteria. A lot of countries have released DSM products for their whole territories with

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resolutions equal or less than 90 m (Chen et al., 2022) and a global DSM product, SoilGrids, is even now available at 250 m resolution (Poggio et al., 2021). However, these DSM products have failed for the most part to provide soil predictions everywhere that are accurate enough to inform decisions at the local level. Indeed, a compilation of the performances of broad-scale DSM products (Fig. 9 in Chen et al., 2022) showed relatively low prediction accuracies for the soil properties included in the GlobalSoilMap specifications (average R^2 between 0.3 and 0.5 across mapped soil properties and soil layers). Furthermore, Lemercier et al. (2022) and Rossiter et al. (2022) observed that the National and Global DSM products were unable to accurately represent the local soil patterns as described by detailed soil maps.

The soil input data used to calibrate the DSM models has been widely recognised as a major limiting factor that causes such inaccurate DSM products (e.g. Loiseau et al., 2021). These data consist of the punctual soil observations and measurements that have been collected over decades of soil surveying around the world and that have been progressively integrated into soil databases (Arrouays et al., 2017). Despite this huge effort, the spatial sets of soil observations are still too sparse to accurately represent the short-range variations in soil properties. Another source of inaccurate soil predictions that is specific to broad scale DSM products has been raised by Brungard et al. (2021). The large geographic areas covered by these products are likely to encompass significant heterogeneity in soil-forming factors and processes, and certainly include areas with fundamentally different soil-landscape-covariate relationships, which may complicate their calibration.

Instead of using the punctual soil observations and measurements for calibrating DSM models, as is usually done in the DSM community, it could be considered as an alternative to use legacy soil maps. Legacy soil maps have already been used as soil input of DSM models in the early stages of DSM activity (Lagacherie et al., 1995, Lagacherie and Holmes, 1997, Bui and Moran, 2003, Grinand et al., 2008). The principle is to extract a set of 'pseudo soil observations' from the soil map by assigning each node of a grid to the soil class in which it is included. The main advantage of such legacy soil data is that the spatial density of the input data is not limited, which may allow local soil patterns to be captured that are usually missed by conventional punctual soil inputs (Rossiter et al., 2022). In fact, provided it is sufficiently detailed (scale $<$ 1:25 000), the soil map contains the soil surveyor's knowledge of the soil landscape relationships gained from the number of borehole and soil surface observations that are not stored in the soil databases (Lagacherie et al., 1995).

However, using soil maps instead of punctual observations also has some disadvantages. The most obvious of these is the uncertainty of the soil predictions derived from these maps, as has been well documented in the past (e.g. Beckett and Webster, 1971; Marsmann and De Gruijter, 1986, Leenhardt et al., 1994). Nevertheless, Leenhardt et al. (1994) observed a significant reduction in uncertainty in texture predictions when going from a 1:100,000 scale to a 1:10,000 scale (the proportion of explained variance increased from 0.30 to 0.69 for sand content and from 0.43 to 0.64 for clay content). Recently, we (Lagacherie et al., 2025) observed moderate uncertainty in soil class predictions derived from a detailed (1:8,000 scale) soil map in India (respective prediction accuracies of 86%, 84%, and 89% for depth, texture, and stoniness). While still non-negligible, the uncertainty affecting detailed soil map inputs cannot therefore be significantly larger than that affecting the measured legacy soil profiles that have classically been used in operational DSM (Arrouays et al., 2014; Poggio et al., 2021) and which have been recently shown to be subject to significant geo-positioning and analytical errors (Lagacherie et al., 2024).

Another disadvantage of using legacy soil maps as input to DSM models is that if only detailed soil maps are used for uncertainty limits, the scarcity of these maps results in very uneven spatial sampling with large unmapped areas to which the soil maps must be extrapolated. There are few attempts in the literature to define extrapolation areas with soil-landscape relationships that are sufficiently similar to those of the soil map perimeter to allow soil map extrapolation. Lagacherie et al. (2001) defined a mathematical distance between soilscapes, where a soilscape represented a high-level grouping of soils based on their relationships to the landscape and was numerically represented by a cover frequency vector of combined soil-forming factor classes. Such a mathematical distance was found to have a good relationship with the representativity of soilscapes, measured as the proportion of soil classes belonging to the list of mapping units of the soilscapes. Later, Wadoux et al. (2024) used the concept of soilscape distances to delineate soil districts across Europe. Bui and Moran (2003) used classification trees to define the most appropriate soil map to extrapolate from for predicting soil classes in unmapped regions. Trees were constructed using each soil map as a separate target class and regions with environmental characteristics most similar to those of the soil map perimeters were identified by applying the decision trees.

In this paper, we aimed to map soil property classes (depth, texture, stoniness) over a large part of the state of Karnataka in South India (156,499 km^2) using as input data 91 detailed soil maps of micro-watersheds, covering a total area of 464 km^2 i.e. 0.3% of the study area. We compared two DSM approaches using such input data: i) a baseline approach, which consisted of building a single model for the entire study region using the set of 91 maps as unique soil input, and ii) an oriented extrapolation approach, which consisted of first selecting the most appropriate map for extrapolation at a given location and then applying the DSM model trained with that map data to that location. The latter procedure used a soilscape distance to select the best map to extrapolate. The two approaches also used such a soilscape distance for ex ante identification of areas with the least uncertain soil class predictions.

2. Material

2.1. Study area

Karnataka, a state in southern India, lies between 11°30′ to 18°30′ N latitude and 74°15′ to78°30′ E longitude. It is divided into four major physiographic regions: the South Deccan Plateau, West Coast Plains, Eastern Ghats, and Western Ghats. This study focuses on the South Deccan Plateau (highlighted in yellow in Fig. 1b).

The South Deccan Plateau is characterized by contrasted soil-forming factors along a north–south gradient. The northern part of the study area has a hot, semi-arid climate, with annual rainfall of 600–750 mm and potential evapotranspiration (PET) of 1600–1700 mm whereas southern part of Karnataka experiences a tropical semi-arid to sub-humid climate, with an average annual rainfall of 700–1,200 mm. The northern part of the region is dominated by rainfed agriculture, with crops such as sorghum, pigeon pea, and pearl millet whereas the southern one include also irrigated areas with paddy and sugarcane. The landscape of the northen part is generally 300–600 m in elevation with occasional hills, and is largely underlain by basalts forming part of the Deccan Traps of Maharashtra whereas the geology of the southern region is primarily granite and granite gneiss.

Such contrasted soil forming factors have generate different pedological contexts within the study area. In the north, the main soils include shallow to deep black soils, red loam, red clay, alluvio-colluvial soils, and lateritic gravelly soils. The soils of the Southern part of Karnataka comprise red gravelly loam, red loam, red gravelly clay, red clay, lateritic soils, and alluvio-colluvial soils.

2.2. Micro-watersheds delineations

The study area was divided into micro-watersheds using natural drainage lines and ridges, in accordance with the National Watershed Atlas framework by Soil and Land use Survey of India (SLUSI, 2012). These boundaries were delineated from topographic maps and high-resolution satellite imagery. First-order streams were mapped and

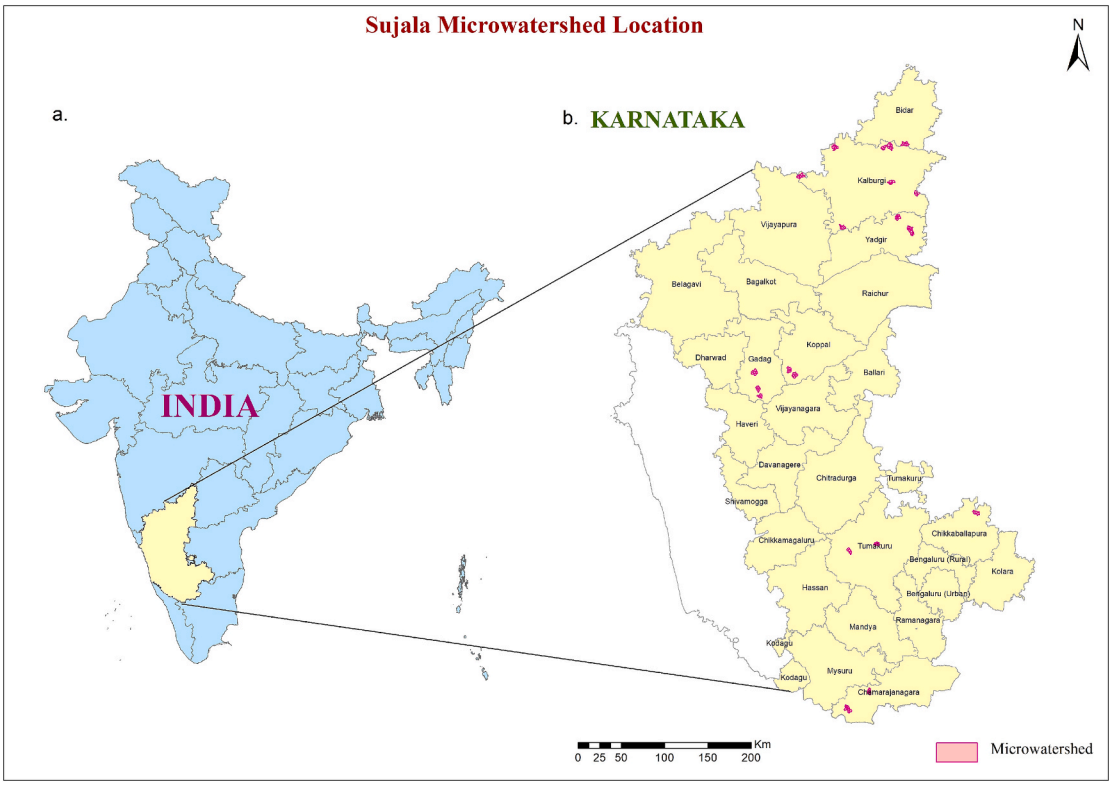


Fig. 1. a) Location of Karnataka state in India, b) Detailed soil maps (in red) in the study area. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

systematically combined to form higher-order streams until a micro-watershed of approximately 500–1,000 ha was defined. Watershed limits were established along ridgelines and elevated terrain separating drainage basins, thereby maintaining hydrological integrity. Each micro-watershed was assigned a unique code under the "All India Watershed Atlas" classification, where larger basins are progressively subdivided into sub-basins, watersheds, and micro-watersheds. The final micro-watershed maps were generated at the cadastral/village scale, serving as a baseline for soil, land use, and resource surveys, and facilitating scientific land resource development and watershed management planning. The state of Karnataka has been divided into 6 rivers basins, 234 watersheds, 3515 sub watesheds and 34,299 micro watersheds (Fig. 2).

2.3. Soil data

The study utilized soil maps of 91 micro-watersheds distributed across eight districts (Chamarajanagara, Tumkur, Chickaballapur, Gadag, Koppal, Yadgir, Kalburgi, and Bidar) and grouped into 19 clusters (figure 1), covering a total area of 464 km^2. The micro-watersheds varied in size from 2.91 to 8.29 km^2, with an average of 5.48 km^2. The soil maps were prepared by ICAR-NBSS&LUP following the procedures described in the field guide (Natarajan et al., 2016) under Sujala III and REWARD projects. These maps were generated at a 1:8,000 scale using high-resolution satellite imagery and cadastral maps with detailed soil profile observations. The soil survey work was conducted by multiple survey teams operating across the study area. To ensure consistent methodology and interpretation standards, all field personnel were centrally trained by scientists from ICAR-NBSS&LUP using the standardized field guide for Land Resource Inventory (LRI), a detailed operational manual developed for soil description, sampling, and mapping under the Sujala III and REWARD projects. Routine quality control and validation by NBSS&LUP scientists were carried out throughout the survey process to maintain uniformity in soil description, classification, and map generation, ensuring that data from different survey teams were consistent and comparable across the study area.

The soils were mapped as phases of soil series and represented on the maps by alphanumeric symbols. For example, the unit BMBmA1g1 corresponds to a phase of the Beemanabeedu (BMB) series with a very deep soil depth class, where the fourth lowercase letter (m) denotes surface soil texture, the fifth uppercase letter (A) indicates slope class, the sixth numeral (1) refers to erosion degree, and the final code (g1) specifies stoniness. Each symbol represents a predefined ordinal class of soil characteristics. Field measurements of soil depth, texture, and stoniness were subsequently categorized into standard classes for interpretation in line with the field guide (Natarajan et al., 2016).

Depth, texture and stoniness classes maps were derived from the genuine soil map by merging polygons with the same soil property classes. These maps were then used to document 57,235 specific locations at the nodes of a regular grid with a 90-metre grid spacing. These datasets were used to calibrate and evaluate the DSM models. The modalities of the considered soil property classes are presented in Table 1.

2.4. Soil covariates

A range of environmental variables was used to predict soil depth, texture and stoniness (Table 2). Topographic attributes including aspect, slope, curvatures, hill shading, relative slope position, total catchment, topographic wetness index (TWI), and valley depth were derived from 30 m SRTM DEM using Saga-GIS 8.4.1. Multi-resolution indices for valley bottom flatness (MrVBF) and ridge top flatness (MrRTF) were included to capture hydrologically significant landforms. Biotic factors were represented using eight Landsat 8 bands (0.43–2.29 μm) along with NDVI and EVI from MODIS to indicate vegetation and land use. Bioclimatic variables, annual precipitation, solar radiation, and temperature extremes were obtained from WorldClim v2. All input layers were reprojected into a common Lambert Conformal Conic projection with the same spatial extent and resampled to a 30 m resolution raster

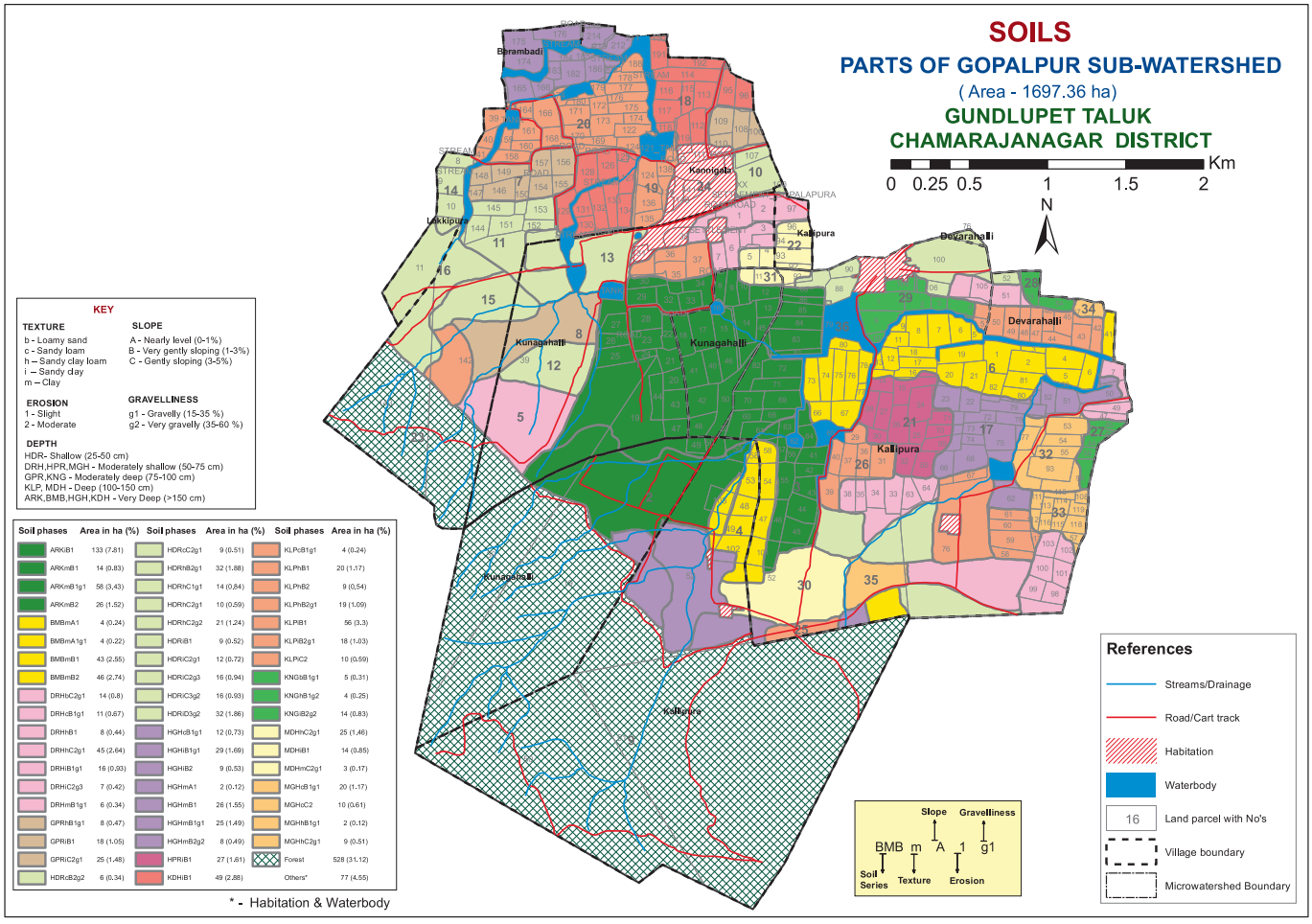


Fig. 2. An example of detailed soil map of micro-watershed (Hegde et al., 2019).

grid to ensure spatial consistency, allowing accurate alignment and covariate extraction across datasets before modelling.

3. Methods

3.1. Predicting soil property classes

A random forest algorithm was used to provide the prediction functions for depth, stoniness and texture classes. The first two soil properties to be predicted were expressed in the soil map as ordinal variables (depth classes and stoniness) with known underlying continuous variables (depth and stone content). The latter, soil texture, is not normally expressed by ordinal classes. However, in this particular case, the soil texture classes could be considered as ordinal because they were ordered according to the clay content (Table 1), which was the underlying continuous variable.

As we did in a previous study using the same type of soil data as input (Lagacherie et al., 2025), we followed the recommendation of Hornung (2019). We converted the three soil properties into continuous variables by assigning the midpoint of the intervals defining their ordinal classes. For depth and stone content, we considered the explicitly defined intervals of quantitative values (Table 1). For texture, we assigned a regular suite of integers (1, 3, 5, 7, 9, 11) to each class to represent the regular increase in clay content across the ordered textural classes presented in Table 1. This allowed a classical regression tree algorithm to be used to construct the prediction functions. Therefore, a quantile forest algorithm (Meinshausen, 2006) was applied using the R package ranger (Wright and Ziegler, 2017). The hyperparameters of this algorithm (mtry, sample fraction and minimum node size) were determined for each soil property class using the procedure available in the TuneRanger package (Probst et al., 2018). The number of trees was set to 500.

Finally, the predicted distributions of the continuous variables were transformed back into the distributions of the originally defined soil properties using a specific R script. This script did the following: i) assign each individual with a predicted continuous value to a given soil property class, using the defined intervals of soil class properties; and ii) count the number of individuals in each predicted class.

Table 1
Soil property classes extracted from soil maps.

Soil depth	texture	Stone
Very shallow (0–25 cm)	Loamy sand	Non stony (0–15%)
Shallow (25–50 cm)	Sandy loam	Stony (15–35%)
Moderately Shallow (50–75 cm)	Sandy clay loam	Very stony (35–60%)
Moderately Deep (75–100 cm)	Clay loam	Extremely stony (60–80%)“)
Deep (100 – 150 cm)	Sandy clay	
Very deep (150 – 200 cm)	Clay	

3.2. Defining a soilscape distance

Following a previous work (Lagacherie et al., 2001), we considered a soilscape to be a landscape unit containing a limited number of soil classes that are geographically distributed according to an identifiable pattern and stable soil-landscape relationships. Such a concept is similar to those defined by other soil scientists around the world, such as pedological provinces (Smeck et al., 1983) or soil combinations

Table 2
The different covariates used in the model.

Covariates	Description	Source	Scorpan factor
Elevation (m)	Height above mean sea level.	SRTM DEM	R
Aspect	Direction a slope faces		
Slope	Steepness of terrain		
Channel network base level	Lowest point in a drainage network		
Channel network distance	Distance to nearest stream/ channel.		
Analytical hill shading	Simulated shading highlighting terrain features.		
MrRTF	Multi Resolution ridge Top Flatness Index.		
MrVBF	Multi Resolution Valley Bottom Flatness Index		
Plan curvature	Horizontal curvature		
Profile curvature	Vertical curvature along slope		
Relative slope	Position along slope		
Convergence index	Tendency of a terrain surface to converge or diverge flow		
TPI	Topographical Position Index (mean difference of elevation with neighbouring pixels)		
TWI	Topographic Wetness index		
Total catchment	Upstream area draining through a point.		
Valley depth	Vertical difference from ridge to valley floor.		
LS factor	Slope length (L) and slope steepness (S)		
Landsat 8 bands		Landsat	O
NDVI	Normalized Difference Vegetation Index	MODIS	
EVI	Enhanced vegetation index		
Bioclimatic variables		WorldClim 2 data	C
Solar radiation			
Precipitation (mm)			
Average ltemperature (°C)			
Maximum temperature (°C)			
Minimum temperature (°C)			
SMU	Soil mapping unit of the 1:250,000 map of Karnataka	Shivaprasad et al., (1998)	S

(Fridland, 1974). Soilscapes have recently been considered as a suitable underlying concept for the delineation of soil districts in Europe (Wadoux et al., 2024). Following this earlier work, we have chosen to describe soilscapes quantitatively by calculating the distribution of soil classes within each micro-watershed, which is considered here as the elementary soil landscape unit.

Based on this quantitative representation of a soilscape, the soilscape distance between two micro-watersheds was defined by i) calculating the Euclidean distances between the depth, texture and stoniness class distributions of two given watersheds and ii) combining the three Euclidean distances thus obtained into a single one by simply averaging them.

4095 soilscape distances between the 91 micro-watersheds with soil maps were calculated using a specific R script. These distances were further considered as the ground-truth soilscape distances.

3.3. Predicting soilscape distances for unmapped micro-watersheds

Soilscape distances between micro-watersheds without a soil map were estimated using a random forest algorithm, using as a learning sample the 91 micro-watersheds with a soil map (Fig. 1) for which 4095 ground-truth soilscape distances were calculated as described above. From the set of available soil covariates, a set of candidate soilscape distances was constructed that could explain the variation in the ground truth soilscape distances. Two types of distance were considered:

- *Distances between distributions of soil covariates.* For quantitative soil covariates, continuous distributions were first converted to quintile distributions, with the quintile threshold being pre-determined once for the entire study area from a large set of 2,480,700 covariate values covering the study area at 250 m resolution. 61 matrices of Euclidean distances between quintile distributions of soil covariates were then computed, generating 4095 x 61 candidate values to explain the ground truth soil distances.
- *Distance between variograms of soil covariates*. Experimental variograms of soil covariates were first calculated within each micro-watershed, considering ten 100 m lags from 0 to 1 km. The distances between variograms were further calculated using Euclidean distances.

In addition to this set of covariates, two other soil covariate distances were added:

- An Euclidian distance between the soil mapping unit distributions of the small scale soil map (see description in the soil covariate section).
- A geographical distance between the 91 micro-watersheds.

The regression random forest algorithm of the R package ranger (Wright and Ziegler, 2017) was trained to select the combination of soil covariate distribution distances that best explains the ground truth. The hyperparameters of this algorithm (mtry, sample fraction and minimum node size) were determined using the procedure available in the TuneRanger package (Probst et al., 2018). The number of trees was set to 500.

3.4. The experiment

Three different approaches to predicting depth, texture and stoniness classes were tested. The first was to build a single DSM model from the entire set of soil-mapped micro-watersheds in the study area. The last two consisted of building 91 different DSM models from each available soil-mapped micro-watershed and, for a given unmapped micro-watershed, applying the DSM model built from the soil-mapped micro-watershed that has the most similar soilscape. This was translated quantitatively by selecting the soil-mapped micro-watershed that had the minimum soilscape distance. The last two approaches differed in the soilscape distances used to select the DSM model to be applied. One considered the ground truth Soilscape distance (section 3.2.) while the other considered the predicted distance (section 3.3.). It should be noted that the first approach is not practically applicable as it would require a soil map of the unmapped micro-watershed. However, we have considered this theoretical approach in order to better understand our results.

The three tested approaches were evaluated using a leave-one-micro-watershed out approach. In each step, one soil-mapped micro-watershed was considered as a validation set. Soil property classes were predicted at each location in this micro-watershed using DSM models calibrated to the 90 remaining soil-mapped micro-watersheds. The leave-one-micro-watershed-out approach was also applied to predict soilscape distances in the third approach, which involved predicted soilscape

distances.

The predictions of the soil property classes were compared with the soil map of the micro-watershed by calculating a prediction accuracy as the percentage of agreement between the soil map at a given location and the predicted soil property class (i.e. the class with the highest probability of occurrence) at the same location. The evaluations of the DSM approach were completed by calculating a prediction uncertainty as the sum of the probabilities of occurrence of the non-predicted classes. Averaged accuracy and uncertainty were calculated for the whole study area by the weighted means of the accuracies and uncertainties calculated at each step of the leave-one-micro-watershed approach, where the weights were the size of the micro-watersheds. To explore the effect of soil landscape distance on the quality of soil class predictions, accuracy and uncertainty were also calculated for different increasing soil landscape distance thresholds.

4. Results

4.1. Predictions of soilscape distances

From the results obtained by the leave-one-micro-watershed-out approach, the Soilscape distances between the 91 micro-watersheds were predicted from covariate distances with a coefficient of determination (R^2) of 0.45. It should be noted that, although the predictions of soilscape distances were globally unbiased (Mean Error of 0.05) the small soilscape distances (< 50) were overestimated, (see Fig. 3).

Fig. 4 shows the 15 most important soil covariates for predicting soilscape distances as calculated by the Ranger method. Covariates related to climate (in blue on the y-axis) and relief (in red) were found to contribute most to the predictions. Geographical distance ("geo" in figure 4) and differences in the distribution of small-scale soil map units within the micro-watersheds ("Usoil" in figure 4) were present among these 15 covariates, but were not by far the most important (9th and 15th). It is worth noting that the covariates representing differences in soil covariate variograms (labelled in red in figure 4) were well-represented among these most important covariates (8 out of 15).

4.2. Comparing DSM overall model performances

The predictions of texture and stoniness classes over the whole study area were fairly accurate and had low uncertainty, whereas the depth classes were much less accurately predicted with higher uncertainty. Predictions were more accurate when a single model calibrated onto the entire set of soil maps was used, as opposed to multiple models, each calibrated onto a single soil map and selected by choosing the one with the minimum soilscape distance. The errors affecting the predicted soilscape distances previously observed had a significant impact on the performance of the multiple model approach, as the prediction accuracy decreased when using predicted rather than ground-truth soilscape distances. However, this impact was less pronounced for depth than for texture and stoniness.

The confusion matrices (see supplementary material S1) provided further insight into the performance of the models. They show that, although some classes are clearly over-represented in the study area (e.g. 32% 'deep soil', 58% 'clayey soil', 59% 'non-stony soils'), they are only moderately over-predicted, and the under-represented classes are not neglected by the models, regardless of the approach. The confusion matrices also reveal that the most mispredicted cases (predicted classes different from more than one class) were only between 28 and 36%, 16 and 27% and between 4 and 10% for the depth, texture and stoniness classes, respectively.

4.3. Prediction performances and soilscape distances

Fig. 5 reveals that, regardless of the soil properties, the prediction accuracy (black curves) decreased significantly for micro-watersheds that were not close to a micro-watershed with a soil map, i.e. that have a large minimum ground-truth soilscape distance from the set of mapped micro-watersheds. For micro-watersheds with a soilscape similar enough to one with a soil map (the left-hand side of Fig. 5

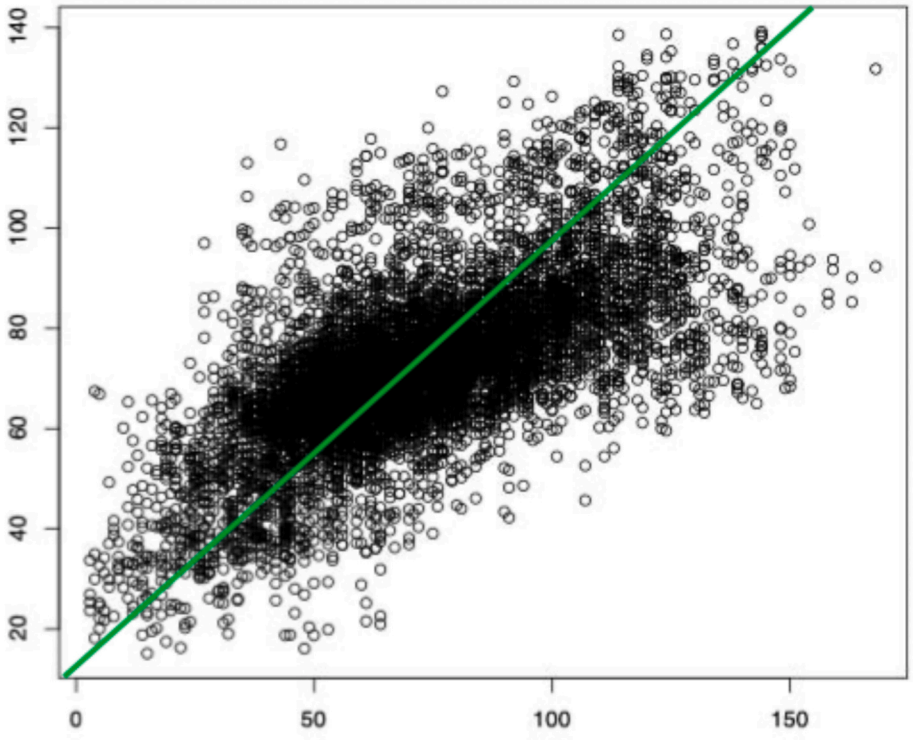


Fig. 3. Ground-truth soilscape distances vs predicted ones (leave one micro-watershed out approach).

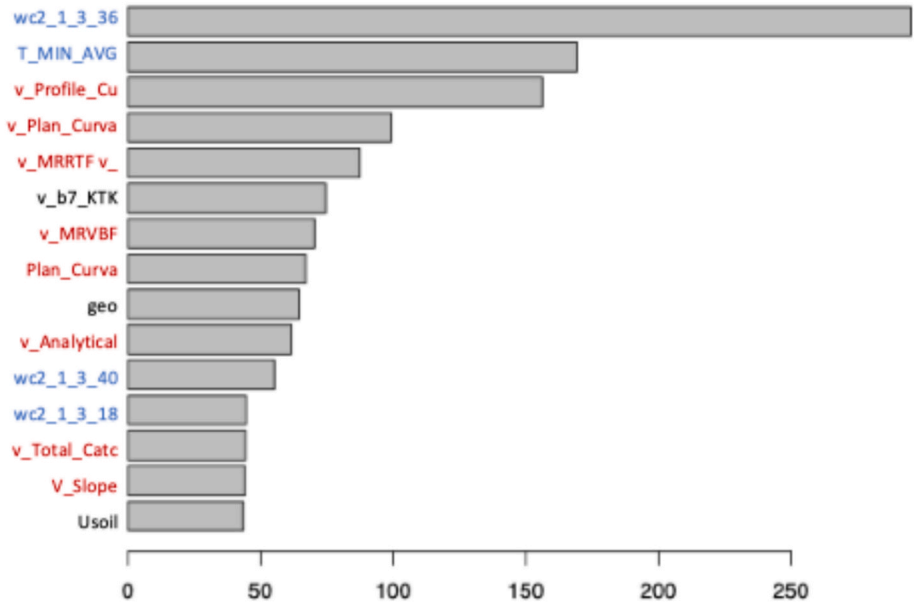


Fig. 4. Importance ratios of the 15 most important soil covariates distribution or variogram distances for predicting soilscape distances between micro-watersheds. V prefix mean variogram distance, y axis labels in blue, red and black are climate relief and other covariates respectively.

Table 3
Overall accuracies and uncertainties (in parenthesis) of soil property class predictions.

	Depth	Texture	Stone
Single model	46% (43%)	73% (26%)	71% (31%)
multiple models with selection by ground-truth soilscape distances	40(44)	69(24)	66 (30)
multiple models with selection by predicted soilscape distances	38(41)	62 (24)	59(28)

graphics), the differences in accuracy between models observed for the entire set of micro-watersheds (two first lines of Table 3) were no longer apparent; both were much higher, especially for texture and stoniness classes.

Inverse evolution was observed for uncertainty in all cases (red curves on figure 5). Whatever the model, there was less uncertainty for micro-watersheds with a soilscape similar enough to one with a soil map. For the predictions of stoniness classes only, the uncertainty was found less when using multiple models..

4.4. Extrapolating soil maps

Fig. 6 shows how the soil class prediction performances evolve when considering the predicted soilscape distances using the model presented in Section 4.1. Unlike Fig. 5, the soilscape distance thresholds were expressed as thresholds of the representative area size here to better visualise the compromises between the size of the predicted area and the prediction performances. This was made possible because the predicted soilscape distances could be computed for the entire set of unmapped micro-watersheds of the study area.

As with Fig. 5, Fig. 6 clearly showed increases in the prediction performance of all soil classes for the smallest representative area sizes (i.e. the smallest predicted soilscape distance thresholds). However, the model performances for all properties reached their minimum values for fairly small representative area sizes (around 20%). Below 20%, the performance decreased regularly, making it impossible to define an optimal threshold that would strike the best balance between performance and the size of the mapped area. As an example, we selected a representative area threshold of 7% of the total area, corresponding to a predicted soilscape distance threshold of 31 (the green, dashed vertical lines in Fig. 6). This represented a medium-term compromise between prediction performance and the size of the mapped area.

Fig. 6 showed that, unlike when ground-truth distances were used (Fig. 5), the single model produced more accurate predictions than the multiple models driven by predicted soilscape distances, even for the smallest representative areas (i.e. the lowest predicted soilscape distance thresholds). This revealed that the loss in accuracy of the soilscape distance predictions affected the selection of the best models required by the multiple models approach. In terms of uncertainty, the multiple model approach continued to perform better for stoniness class predictions, despite using predicted soilscape distances.

To map an example of representative area (Fig. 7), the micro-watersheds having a minimum soilscape distances from a mapped micro-watershed below the above defined threshold (31), were included in the extrapolation area of the entire set of soil maps. As already shown on figure 6, the accuracy and uncertainty of all the soil class predictions (Table 4) were higher and lower, respectively, than previously observed across the entire study area (Table 3).

Fig. 7 shows the extrapolation area, which is 10,892 km^2, representing 7% of the study area. This was 21 times larger than the total area of the mapped micro-watersheds. This area comprised large clusters located in districts with the highest number of mapped micro-watersheds (Belgavi and Yagdir in the north, Gadag in the centre, and Chamarajanagar in the south) but also in districts that did not contain any mapped micro-watersheds (Mandya, Mysore, Kodagu and Raichur). Furthermore, extrapolation of a mapped micro-watershed was also possible in all the other districts of the study area, albeit in small sets of isolated micro-watersheds. Finally, It is also important to note that the representative area is more important in the northern part of the study area, characterized by more uniform patterns of black soils derived from basaltic parent material,

The 91 mapped micro-watersheds exhibited moderate variations in the area to which they could be extrapolated (Fig. 8). For most of these micro-watersheds (80%) the extrapolation areas did not exceed 200 km^2. However, five micro-watersheds had a more than 500 km^2 representative area among which the micro-watershed named Balbatti5, located in the northern part of the region, reached a representative area of 1,245 km^2 representing alone 5.5% of the total extrapolation area.

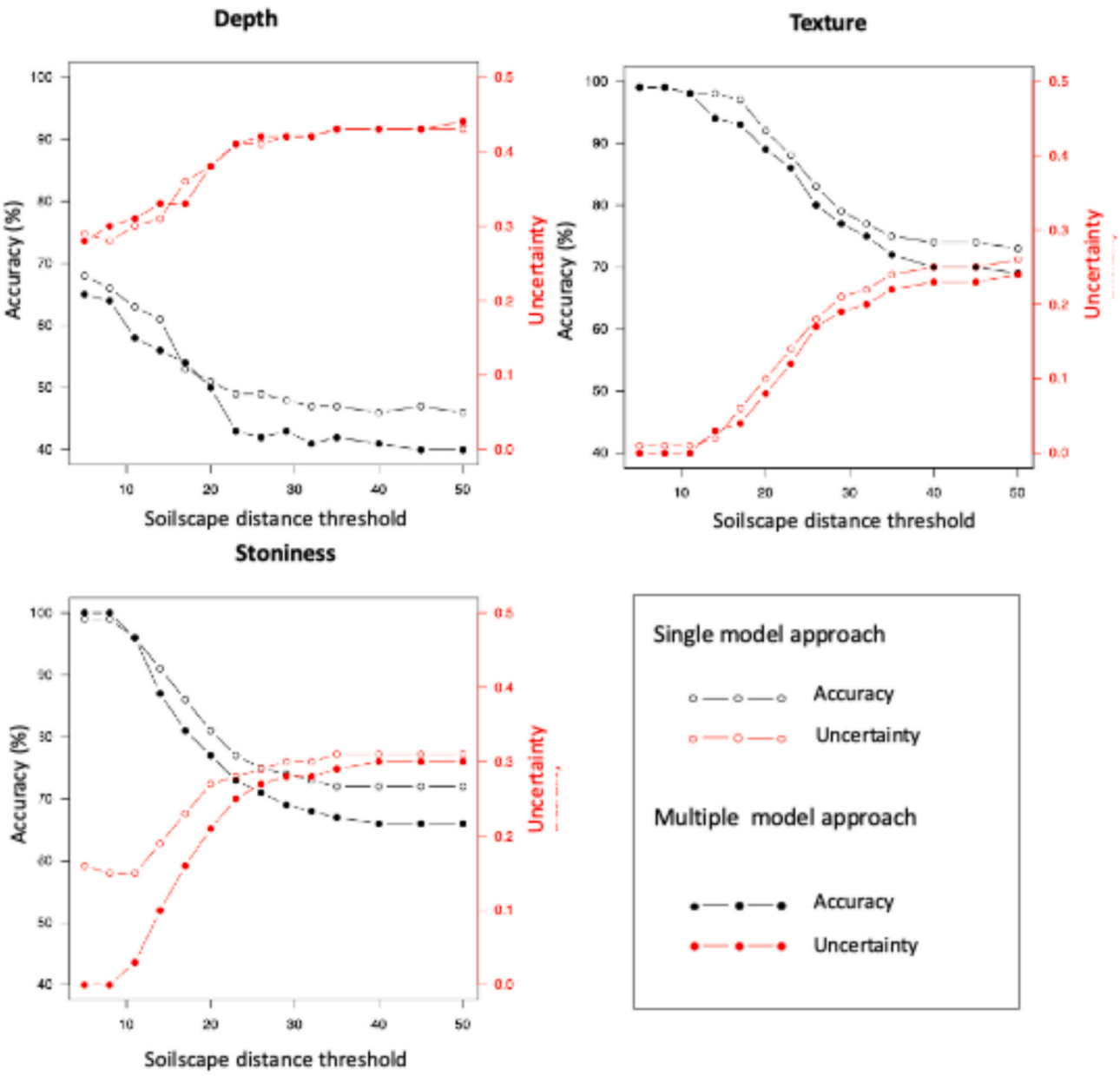


Fig. 5. Accuracy and uncertainty of soil property class predictions for different ground-truth soilscape distance thresholds using either a single model or multiple models.

5. Discussion

5.1. Using legacy soil maps as DSM model inputs

The use of legacy soil maps as input for DSM models to map soil properties has rarely been considered in the literature. To the best of our knowledge, only Bui and Moran (2003) deviated from the common practice of using legacy site soil observations and measurements to produce broad scale DSM products (Arrouays et al., 2014; Chen et al., 2022). However, their approach in the Murray-Darling Basin in Australia involved filling the gaps between a small number of soil maps at various scales that already covered most of the study area. This differs greatly from our attempt to extrapolate from very detailed soil maps that together cover only a tiny part of the study area (0.33%).

The performance of the single model's predictions, when using the entire set of soil maps, can be compared with other attempts to map soil property classes over areas of similar size. In terms of texture class mapping, Dharumarajan and Hegde (2022) in the neighbouring Indian state of Andhra Pradesh, Mallah et al. (2022) in Iran's Guilan Province, and Eymard et al. (2024) in France's Loiret Department obtained overall accuracies of 52%, 53%, and 49%, respectively. These results are much lower than those observed in our study, where the accuracy was 73% for the whole study area. Conversely, Brungard et al. (2021) in the western US achieved a higher level of accuracy in predicting soil depth classes (62% vs. 46%). Beyond the specifics of each study area, we interpret these contrasting results as being due to the greater or lesser uncertainty of the detailed soil maps, which either cancelled out or did not cancel out the benefits of dense spatial sampling and filtering out very short-scale unmappable variability.

In the case of mapping soil depth classes specifically, the poorer predictive performance should be interpreted as resulting from weaker relationships with the soil covariates, particularly the remote sensing ones, which only characterise soil surfaces.

Additionally, it should be noted that we used a set of recent soil maps, all of which were developed by the same team using the same well-defined approach (section 2.3.), thereby limiting discrepancies

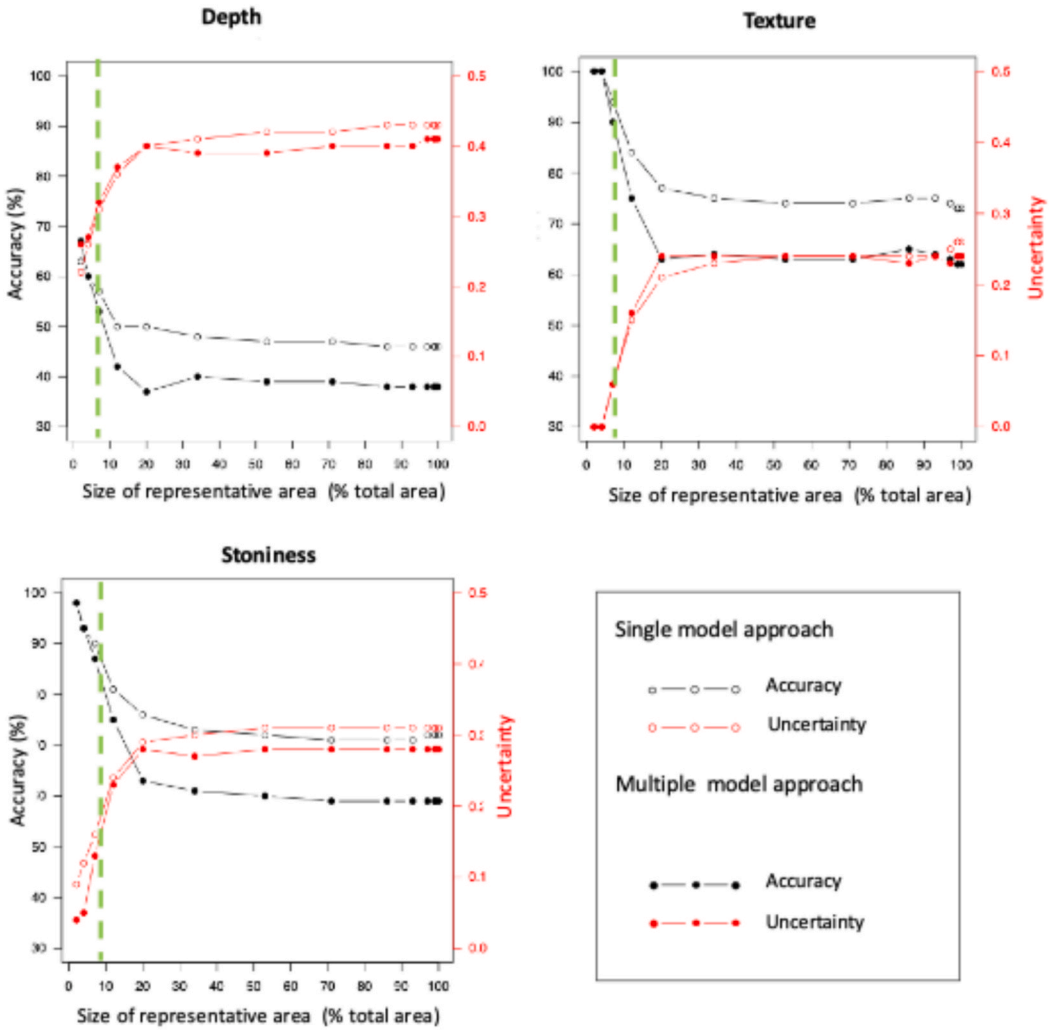


Fig. 6. Accuracy and uncertainty of soil property class predictions using either a single model or multiple models for different sizes of representative areas obtained by thresholding the predicted soilscape distances. Vertical green dashed line represent the threshold of size (7% of the total area) selected as example for mapping a representative area.

Table 4
Accuracies and uncertainties (in parenthesis) of soil property class predictions for the extrapolation area (threshold of soilscape distance at 31).

	Depth	Texture	Stone
Single model	57% (31%)	94% (6%)	90% (7%)
multiple models with selection by soilscape distances	53% (32%)	90% (6%)	87% (13%)

across maps. This favourable condition probably increased the performance of the DSM models calibrated onto soil maps against those calibrated onto a classical set of legacy soil maps. Therefore, it can be concluded that using detailed soil maps to calibrate DSM models can yield competitive accuracies for some properties, provided that the map quality and consistency are favourable.

5.2. Soilscape distance for DSM

The presented approach is one of the rare DSM approaches that explicitly considers one upper level of organisation of the soil cover. This level is referred to here as 'soilscape', although it has been widely reported by soil scientists around the world under different terms (Astle et al., 1969; Fridland, 1974; Favrot, 1981; Smeck et al., 1983). This level has also been used as the basis for defining the mapping units of conventional small-scale soil maps (Bornand et al., 1989; Costantini et al., 2012). Previously, Brungard et al. (2021) tested an ensemble of individual DSM models, each calibrated within a well-defined eco-physiographic region, as an alternative to the traditional single model calibrated across the entire mapped area. This assumes that each eco-physiographic region has specific soil-landscape relationships. Wadoux et al. (2024) used the concept of soilscapes to map soil districts in Europe, aggregating soilscapes quantitatively represented as distributions of continental-scale soil units or soil covariates.

Building on previous work in this area, we considered the quantitative soilscape distance between elementary geographical units, defined here as micro-watersheds. This distance could be predicted from accessible soil covariates and used as a novel input in a DSM approach. Initially, we conceived such a soilscape distance as a means of selecting the most suitable model to apply to a given micro-watershed, based on soilscape similarity between mapped and unmapped micro-watersheds. However, our results revealed that the soilscape distances could also be used to evaluate the uncertainty of soil class predictions in advance whatever the DSM model used. This enabled us to delineate representative areas within the study area characterised by micro-watersheds with small soilscape distances from the soil maps. Within these defined representative areas, the soil class predictions were much more accurate and much less uncertain than elsewhere because sharing the

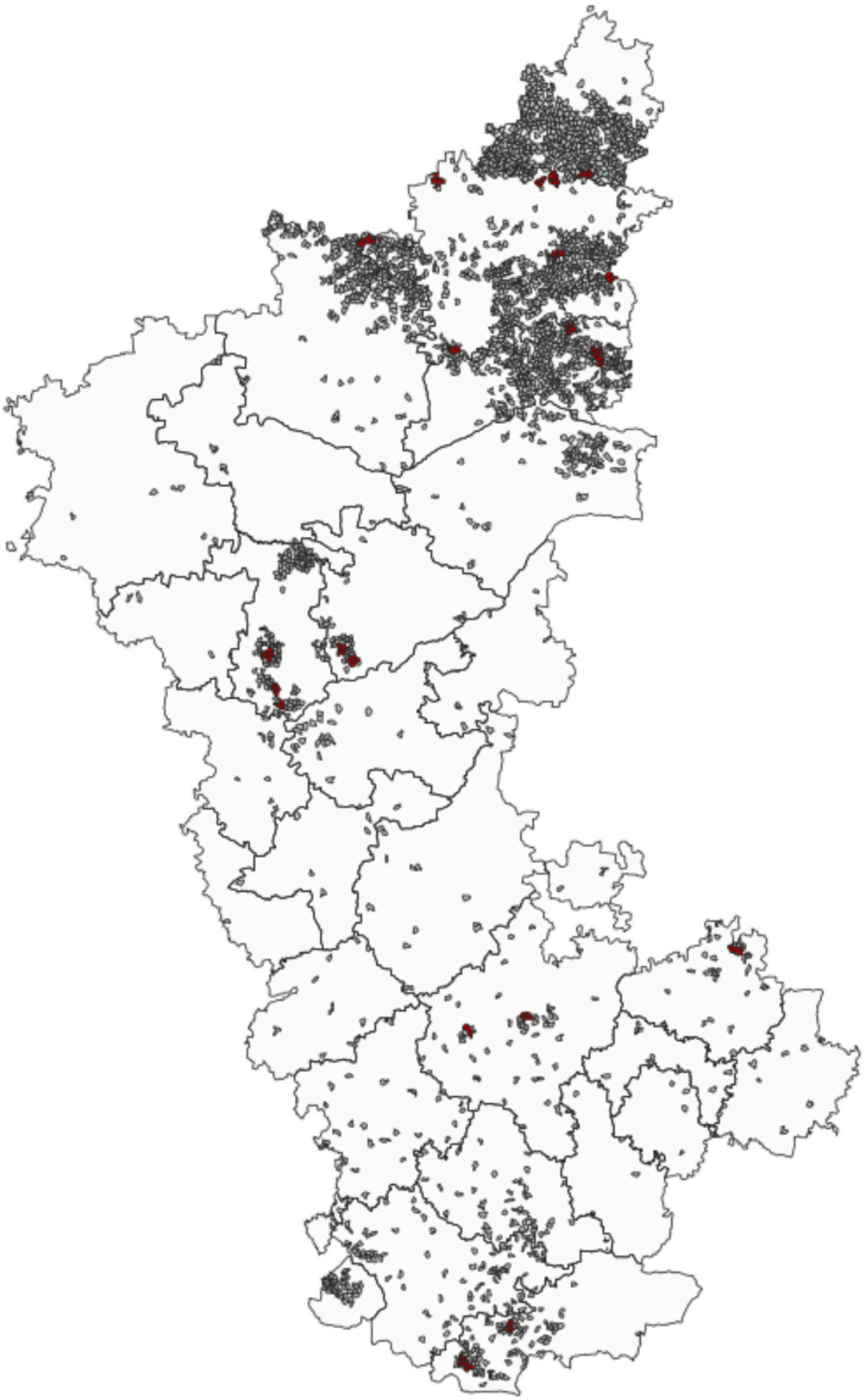


Fig. 7. Extrapolation area (in grey) defined by a soilscape distance threshold of 31. The mapped micro-watersheds are in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

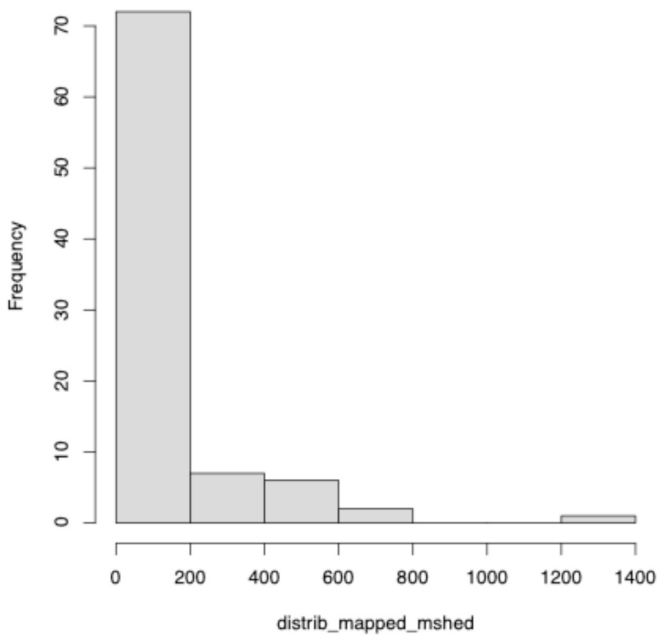


Fig. 8. Distribution of the extrapolation area sizes over the 91 mapped micro-watersheds.

same soilscape (Table 4). This encouraging result paves the way for more efficient DSM approaches that explicitly consider the upper levels of soil cover organisation, which are familiar to soil surveyors but have not yet been incorporated into DSM approaches. Furthermore, the clear relationship between soilscape distances and DSM model performance (Figs. 5 and 6) suggests that more accurate ex-ante estimations of DSM model uncertainty are possible when such soilscapes are considered. In particular, incorporating soilscapes could mitigate the systematic underestimation of uncertainty observed in traditional DSM methods, especially in cases of heavily clustered spatial sampling (Lagacherie et al., 2020), which is prevalent in operational DSM applications.

5.3. Single model or multiple model

In this study, we compared the performance of two adverse DSM approaches: using all the available soil maps to build a single model, or using each soil map separately to build multiple models and then selecting the correct model based on the soilscape distance between the micro-watershed with the soil map and the micro-watershed to be mapped.

Overall prediction performance (Table 3) showed that the single model approach clearly outperformed the multiple model approach. However, when focusing on micro-watersheds closest to those with soil maps (Fig. 4), the two approaches performed similarly (Fig. 5). Nevertheless, inaccurate predictions of soilscape distances (Fig. 3) restored the superiority of the single model approach, even for the closest micro-watersheds (Fig. 6).

In the future, it could be interesting to consider an intermediate approach between those tested in this paper. Following Brungard et al. (2021), this approach would consist of building an ensemble model based on a combination of weighted single models. These weights could then be determined using the soilscape distance probabilities predicted by the random forest.

5.4. Practical outcomes for Karnataka soil mapping (and elsewhere)

This study builds on past and ongoing soil inventory programmes that have, until now, relied solely on detailed, traditional soil surveys of micro-watersheds. We have paved the way for a new approach for mapping soils over large territories with high precision while maintaining acceptable time and cost constraints. This is particularly true for soil properties that can be mapped in detail and are well-correlated with soil covariates, such as texture and stoniness class, as demonstrated in this study. This hybrid approach would combine conventional surveying over limited areas with a DSM approach for extrapolating the detailed soil maps to the rest of the territory. In this perspective, we demonstrated in the Karnataka state that a small number of mapped micro-watersheds (91), representing a small cumulative area (464 km^2), could be extrapolated to cover a much larger area (10,892 km^2) using DSM. It is important to note that this result was obtained after choosing a compromise between mapping accuracy and the size of the extrapolation area (i.e. the soilscape distance threshold was set to 31). Reducing the soil scape distance would have produced more accurate soil class maps, but would have decreased the extrapolation area. In the absence of any objective criteria that can be agreed upon, it is up to the end user to decide which compromise best suits their needs.

The occurrence of a few mapped micro-watersheds with greater extrapolation areas than the majority (Fig. 8) suggests that the selection of micro-watersheds should be more rigorous to maximise the extrapolation-to-mapped area ratio. A numerical sampling method based on the soilscape distance defined in this study could be envisaged in the future, which would deepen the hybridisation between DSM and conventional soil surveying.

Our approach could be applied to other parts of the world where detailed soil surveys have been or are being undertaken at a regional or state level. Examples include the French 'secteurs de référence' (Favrot, 1981). Other options concerning the type of machine learning algorithm or the preparation of the covariates related to soilscape distances will also need to be tested to confirm and improve the presented results. We are eager to see how the local pedological contexts of other places around the world will impact the results of the proposed approach, either positively or negatively.

6. Conclusions

The main lessons we should retain from this study are the following:

- Provided they are detailed enough, soil maps can efficiently be used to calibrate DSM models as an alternative to classical legacy sets of soil profiles. This is a way of coping with the low spatial densities of the latter, which hinder the capture of local soil variation patterns.
- Extrapolation areas can be identified around soil maps using soilscape distances, which provide a numerical account of higher levels of organisation in soil cover that have been observed by a lot of soil surveyors across the world.
- Such soilscape distances have been found to be useful in DSM approaches, both for refining ex ante estimation of soil prediction accuracy and for selecting the appropriate soil map(s) for calibrating DSM models. However, improving the estimation of soilscape distances from landscape covariates is required for the latter task.
- In our study area, we observed that a significant part of Karnataka territory could be covered by accurate maps of texture and stoniness classes using a detailed conventional survey covering a limited area. This paves the way for the integration of conventional soil surveys and digital soil mapping in the operational programmes of soil inventories at state level.
- Further work is needed to refine distance thresholds, evaluate alternative ensemble weighting schemes and integrate soilscape-based sampling design.

CRediT authorship contribution statement

Philippe Lagacherie: Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **S. Dharumarajan:** Writing – review & editing, Validation, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Lagacherie reports administrative support, article publishing charges, and travel were provided by French National Institute for Agricultural Research INRAE. Lagacherie reports a relationship with French National Institute for Agricultural Research INRAE that includes: employment. Lagacherie has patent pending to. Nothing to declare If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.geoderma.2026.117743.

Data availability

The authors do not have permission to share data.

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